Assimilation of *MetOp-C* AMSU-A Data Using the ARMS as an Observation Operator in the YH4DVAR System

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ABSTRACT

The Advanced Radiative Transfer Modeling System (ARMS), a computationally efficient satellite observation operator, has been successfully integrated into the YinHe four-dimensional variational data assimilation (YH4DVAR) system. This study investigates the impacts of assimilating Advanced Microwave Sounding Unit-A (AMSU-A) observations from the *Meteorological Operational Satellite-C (MetOp-C)* on the performance of YH4DVAR. Through a month-long global statistical analysis and a case study of Typhoon Hinnamnor, we evaluate the benefits of AMSU-A data assimilation under clear sky conditions. Key findings are as follows. (1) ARMS achieves simulation accuracy comparable to RTTOV (Radiative Transfer for the Television and InfraRed Observation Satellite Operational Vertical sounder) version 11.2, demonstrating only a 0.5% discrepancy in data retention after quality control. (2) Implementation of ARMS as an operator in YH4DVAR enhances forecast accuracy for the 850-hPa temperature and 500-hPa geopotential height in the tropical region. (3) Compared to RTTOV, ARMS has improved the intensity forecast of Typhoon Hinnamnor and reduced mean wind speed errors by approximately 2% and central pressure errors by approximately 1%. ARMS has now been operationally adopted as an alternative observational operator within YH4DVAR, demonstrating exceptional numerical stability, computational efficiency, and promising potential for future satellite data assimilation applications.

Key words: Advanced Radiative Transfer Modeling System (ARMS), Radiative Transfer for the Television and Infrared Observation Satellite Operational Vertical Sounder (RTTOV), data assimilation, Advanced Microwave Sounding Unit-A (AMSU-A)

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1. Introduction

Currently, meteorological satellite data constitute more than 90% of the inputs in data assimilation systems, provide more accurate initial conditions for numerical weather prediction (Bauer et al., 2006; Le Marshall et al., 2006; Collard and McNally, 2009; Geer et al., 2017), and significantly increase the scores of global and mesoscale forecasts (Le Marshall et al., 2004). These data have also extended the scientific understanding of extreme weather and climate such as typhoons (Zou et al., 2013) and heavy rainfall (Olsen et al., 2015). It should be noted that the observed radiance data from satellites rather than conventional variables (such as temperature, pressure, humidity, and wind) are used in numerical weather predication (NWP). Consequently, direct assimilation of satellite radiance data can only be achieved through the application of radiative transfer models. Therefore, a fast and accurate radiative transfer model has become a core component of satellite data assimilation systems.

For brevity, radiative transfer models are broadly categorized into physical, fast, and artificial intelligence (AI) models. In particular, physical models calculate atmospheric absorption, scattering effects induced by clouds and precipitation, as well as surface radiation through the implementation of line-by-line (LBL) trans-

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mittance models coupled with advanced scattering theory (Buehler et al., 2005; Eriksson et al., 2011; Alvarado et al., 2013; Schreier et al., 2014). The advantage of physical models is their ability to simulate atmospheric transmittance accurately under clear-sky conditions. However, these methods require substantial computational time, which does not meet the timeliness requirements for NWP.

In contrast to physical models, fast radiative transfer models employ statistical regression techniques to establish empirical relationships between forecast variables and atmospheric transmittance to simulate satellite radiance. Notably, these advanced models achieve a remarkable balance between computational efficiency and physical accuracy, demonstrating substantial reductions in central processing unit (CPU) time while maintaining rigorous precision standards. Currently, the most common fast radiative transfer models are the Radiative Transfer for the Television and Infrared Operational Satellite (TIROS) Operational Vertical Sounder (RTTOV) model, developed by the Satellite Application Facility on Numerical Weather Prediction (NWP SAF) (Matricardi et al., 2004; Saunders et al., 2018), and the Community Radiative Transfer Model (CRTM), developed by the U.S. Joint Center for Satellite Data Assimilation (JCSDA) (Weng and Liu, 2003; Weng, 2007). Both models have been widely integrated into data assimilation systems. For example, the weather research and forecasting model data assimilation (WRFDA) system supports both the RTTOV and the CRTM (He, 2018), and the grid point statistical interpolation (GSI) system utilizes the CRTM (Wei et al., 2022). The ECMWF Integrated Forecast System (IFS) (Matricardi et al., 2004) and the Yinhe four-dimensional variational data assimilation (YH4DVAR) system of the National University of Defense Technology (Ma et al., 2022) support the RTTOV fast radiative transfer model.

As an alternative approach, AI-enhanced radiative transfer models leverage machine learning techniques to simulate atmospheric radiation, showing considerable promise in direct modeling applications (Chevallier et al., 2000; Ukkonen, 2022). However, they lack essential tangent-linear and adjoint models, preventing their integration into operational assimilation systems.

Since 1988, China has launched more than a dozen polar and geostationary meteorological satellites. Despite this significant achievement, the optimal utilization of these satellite data in NWP systems remains suboptimal, primarily due to the lack of a dedicated fast radiative transfer model specifically designed for Chinese meteorological satellite instruments (Yang et al., 2020). To address this shortcoming, China Meteorological Administration (CMA) has developed a novel radiative transfer model known as the Advanced Radiative Transfer Modeling System (ARMS). Unlike the traditional CRTM and RTTOV systems, the ARMS offers unique features. In terms of radiative transfer solutions, the ARMS supports various algorithms, including the polarized two-stream approximation (Liu and Weng, 2002), the advanced doubling-adding (ADA) method (Liu and Weng, 2006), the vector doubling-adding (VDA) method (Evans and Stephens, 2010), and the vector discrete ordinate radiative transfer model (VDISORT) (Weng, 1992). In contrast, the CRTM and RTTOV methods primarily utilize ADA and delta-Eddington approximation (Kummerow, 1993), respectively, as their main solutions. In terms of polarization, whereas the RTTOV and the CRTM consider only surface polarization characteristics (Yang et al., 2020), the ARMS accounts for both cloud and surface polarization features, endowing the ARMS with a distinctive ability to simulate cloud scattering.

Currently, the ARMS has been extensively implemented across various domains of meteorological satellite remote sensing applications. It was comprehensively introduced by Weng et al. (2020) regarding its computational processes, simulation capabilities, and future development directions, and Yang et al. (2020) discussed ARMS competencies in satellite assimilation and remote sensing applications. In terms of technical upgrades, Shi et al. (2021) proposed a novel solver called the discrete ordinate adding method, which enhances the radiative transfer computation capabilities of the ARMS. Moreover, Kan et al. (2020) developed fast transmittance coefficients for an ARMS on the *Fengyun-4A*/Geostationary Interferometric Infrared Sounder (*FY-4A*/GIIRS) platform. In applied research, Tang et al. (2021) utilized the ARMS to characterize biases in the *FY-4A*/Advanced Geosynchronous Radiation Imager (*FY-4A*/AGRI) infrared band, demonstrating its practicality in environmental applications. For the inversion of typhoon thermal structures, the use of the ARMS within a 1DVAR framework has been reported to achieve higher accuracy than the CRTM, with the potential to reduce humidity profile errors by 5% (Hu and Weng, 2022). Furthermore, It is reported that China Meteorological Administration global forecast system (CMA-GFS) now supports the ARMS and has implemented its operational applications.

To evaluate the performance of the ARMS within the YH4DVAR assimilation framework, a comprehensive evaluation is conducted through the assimilation of Advanced Microwave Sounding Unit-A (AMSU-A) data from the *Meteorological Operational Satellite-C (Met-*

Op-C). The assessment methodology incorporates both long-term statistical analysis and a case study of Typhoon Hinnamnor (2022), with particular emphasis on multiple performance metrics including: spatiotemporal characteristics of observation-minus-background (O − B) differences, bias correction, quality control, innovation of assimilation, forecast skill, and computational efficiency. This multi-faceted approach ensures a rigorous and comprehensive evaluation of the ARMS system's operational capabilities.

The structure of this paper is as follows. Section 2 details the characteristics of the *MetOp-C* AMSU-A instrument and the four-dimensional variational data assimilation (4DVAR) system based on the ARMS. Section 3 presents the assessment analysis following the integration of the ARMS into YH4DVAR. Section 4 provides a summary of the paper and an outlook for future ARMS applications.

2. Data and methods

2.1 *Advanced Microwave Sounding Unit-A (AMSU-A)*

The Advanced Microwave Sounding Unit-A (AMSU-A) is a 15-channel cross-track scanning total power radiometer. Table 1 lists main characteristics of the AMSU-A channels. Atmospheric temperature profiles are primarily based on measurements obtained at Channels 3–14 near 60 GHz, which is an oxygen absorption band. In addition, several AMSU-A window channels at frequencies of 23.8, 31.4 and 89 GHz are utilized to determine the cloud liquid and ice water contents because they directly respond to emission from liquid droplets and scattering from ice particles. AMSU-A employs a cross-track scanning mechanism, with each scan line consisting of 30 observations at a nadir resolution of 48 km and a swath width of approximately 2343 km. The scan angle at the edge of the orbit is 48.33° (Yan et al., 2020).

Since the launch of the first AMSU-A in 1998, its data have become the most important satellite observation data sources for NWP (English et al., 2000). To comprehensively evaluate the assimilation performance and forecast skill of the ARMS within the YH4DVAR, this study utilizes AMSU-A observations from the *MetOp-C* platform as the primary experimental dataset. The selection of *MetOp-C* AMSU-A instrument for this paper is based on two principal scientific reasons. First, numerous studies have consistently demonstrated the significant positive impact of AMSU-A data on assimilation processes, establishing it as an optimal reference for model evaluation. Second, as the most recently launched platform in the MetOp series, *MetOp-C* exhibits superior instrument stability compared to its predecessors (*MetOp-A* and *MetOp-B*), thereby ensuring higher data quality and reliability for our experimental analysis. This selection not only enhances the credibility of our assessment but also establishes a robust operational framework for evaluating the ARMS model's performance.

2.2 *Four-dimensional variational data assimilation based on the ARMS and RTTOV*

Within the YH4DVAR assimilation framework, both ARMS and RTTOV are implemented as optional observation operators to facilitate efficient assimilation of satellite observational data. The system execution begins with the selection of either RTTOV or ARMS as the primary radiative transfer model. After model selection, the system initiates the forward model computation to simulate satellite brightness temperatures, accompanied by rigorous quality control procedures to eliminate anomalous observations. Upon successful quality control validation, the system iteratively executes the tangent-linear and adjoint models of the selected observation operator to minimize the cost function through variational adjustment of the background field. This iterative optimization process continues until the background field con-

Table 1. Characteristics of AMSU-A channels

Channel index	Center frequency (GHz)	Polarization	Peak energy contribution height (hPa)	Main purpose
1	23.800	QV	Surface	Clouds & precipitation
2	31.400	QV	Surface	Clouds & precipitation
3	50.300	QV	Surface	Clouds & precipitation
4	52.800	QV	1000	Temperature
5	53.596 ± 0.115	QH	700	Temperature
6	54.400	QH	400	Temperature
7	54.940	QV	270	Temperature
8	55.500	QH	180	Temperature
9	57.290 (f0)	QH	90	Temperature
10	f0 ± 0.217	QH	50	Temperature
11	f0 ± 0.322 ± 0.048	QH	25	Temperature
12	f0 ± 0.322 ± 0.022	QH	12	Temperature
13	f0 ± 0.322 ± 0.010	QH	5	Temperature
14	f0 ± 0.322 ± 0.004	QH	2	Temperature
15	89.000	QV	Surface	Clouds & precipitation

Note: QV: quasi-vertical and QH: quasi-horizontal.

verges to an optimal representation of the true atmospheric state. A schematic of the YH4DVAR system is presented in Fig. 1.

2.2.1 *YH4DVAR*

The YH4DVAR system is an operational system that employs a global spectral model (YHGSM) to enforce a dynamic balance during the assimilation process, utilizing a 12-h assimilation window. This approach enhances the accuracy of the initial atmospheric conditions (Zhang et al., 2012). In the YH4DVAR system, observations from the AMSU-A, Microwave Humidity Sounder (MHS), Microwave Temperature Sounder (MWTS), Microwave Humidity Sounder (MWHS), and Global Navigation Satellite System (GNSS) radio occultation bending angle data have already been assimilated, yielding significant improvements. In the context of the fast radiative transfer (RT) model, RTTOV version 11.2 (RTTOV11.2) was utilized in the YH4DVAR system to simulate the satellite radiance. (Ma et al., 2022). The YHGSM adopts a nonhydrostatic moist global spectral dynamic core with an dry-mass vertical coordinate (Peng et al., 2019) and the fast spherical harmonic transformation algorithm (Yin et al., 2018). In the YHGSM, the spherical harmonic spectrum is expanded using a triangular truncation method, with a truncated wavenumber of 1279. The model utilizes a linear reduced Gaussian grid, characterized by an approximate grid spacing of 16 km. It encompasses 137 vertical levels, extending from the earth's surface up to the 0.01 hPa level in the atmosphere. The temporal resolution is set at 600-second intervals.

2.2.2 *ARMS 1.2 and RTTOV11.2*

The ARMS is a fast model for computing the radiances at the top of the atmosphere or at any altitude observed by remote sensing instruments. Its current version (ARMS1.2) supports accurate simulations of satellite observations of atmospheric sounders and imagers from microwave to infrared wavelengths launched by international space agencies or commercial sectors (Weng et al., 2020). It has been operationally implemented in CMA global forecast systems (CMA-GFS) to accelerate the assimilation of Fengyun satellite data and has resulted in positive impacts on global medium-range forecasts.

The ARMS software is developed with modular designs with a plug–play function and allows for user communities to test and implement their own modules in radiative transfer processes. Its atmospheric transmittance module is parameterized with a set of predictors related to atmospheric and surface variables and instrument viewing angles. Various sets of look-up tables (LUTs) have been developed for fast and accurate calculations of scattering and absorption parameters from molecules, aerosols, clouds and precipitation. The surface emissivity is derived from physical models over oceans and from databases pertaining to land.

RTTOV11.2 is a fast radiative transfer model developed by the NWP SAF. It is designed to simulate top-of-atmosphere radiances from passive visible, infrared, and microwave downwards-viewing satellite instruments, such as radiometers, spectrometers, and interferometers (Saunders et al., 2018). The core of RTTOV is a fast parameterization of layer optical depths due to gas absorption. It also optionally computes the Jacobian matrix, which describes the change in radiance for a change in any element of the state vector, assuming a linear rela-

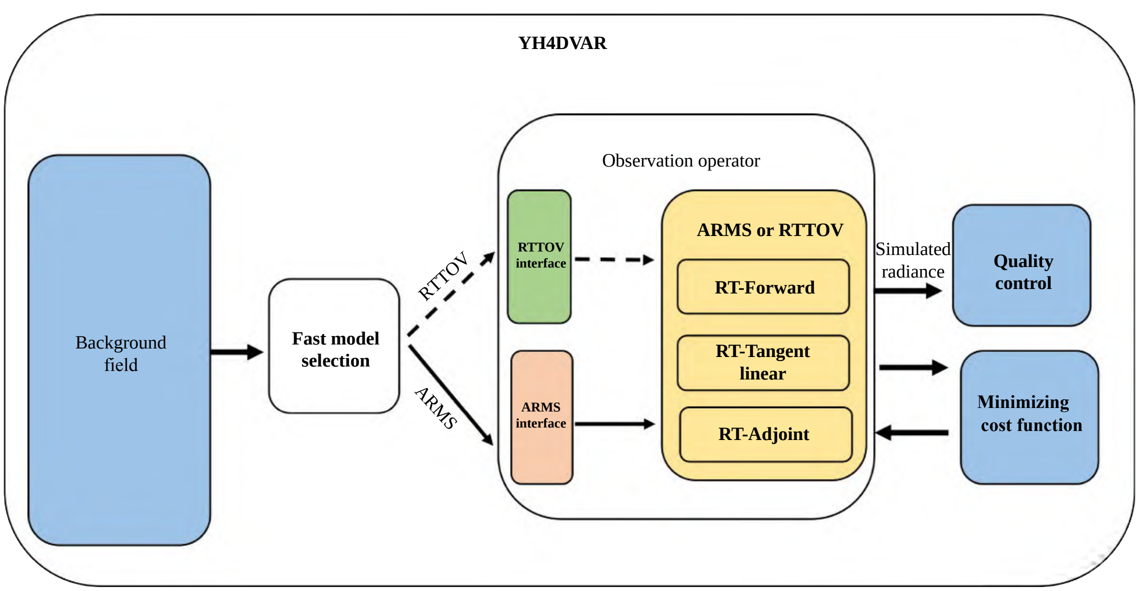

Fig. 1. Schematic diagram of YH4DVAR.

tionship about a given atmospheric state. This feature makes RTTOV particularly useful for data assimilation and retrieval applications (Lupu and Geer, 2015). The primary distinctions between the ARMS and RTTOV in simulating AMSU-A channel brightness temperatures are detailed in Table 2.

2.2.3 *ARMS interface in the YH4DVAR system*

The YH4DVAR system generates atmospheric profiles that are incompatible with the ARMS radiative transfer model's input requirements, particularly in unit conventions and data dimensionality. To address this issue, the atmospheric data must undergo preprocessing to meet ARMS interface specifications. The atmospheric profile data generated by the YH4DVAR system are structured according to a pressure level (P_Level) scheme, which fundamentally differs from the pressure layer (P_Layer) based storage format required by the ARMS interface specification. This structural discrepancy in atmospheric data representation between the two systems is illustrated schematically in Fig. 2.

3. Experimental design and results

This study utilizes AMSU-A data from the *MetOp-C* platform to evaluate the performance of the ARMS within the YH4DVAR system. The selection of AMSU-A data is based on their widespread adoption in meteorological observation systems, supported by numerous studies demonstrating their significant positive impact on assimilation and forecast accuracy when processed through RTTOV observation operator (Bormann et al., 2013). AMSU-A data are assimilated within the YH4DVAR system, and a comparative analysis of the assimilation and forecast effects between the ARMS and RTTOV observation operators is conducted to assess the performance of the ARMS.

3.1 *Experimental design*

In this study, we implemented two experimental configurations to evaluate the performance of different observation operators within the YH4DVAR framework. Both experimental setups assimilated conventional observational data alongside AMSU-A satellite radiance data, with the critical distinction being the choice of radiative transfer model: the ARMS1.2 in one configuration and the RTTOV11.2 in the other. Specifically, the experimental design comprises Test_arms, which utilizes the ARMS1.2 forward operator and its associated tangent-linear and adjoint models, and Test_rttov, which employs the corresponding RTTOV11.2 components. The complete experimental configuration, including detailed parameter specifications, is presented in Table 3. The evaluation period encompasses the entire month of August 2022, during which assimilation cycles were executed twice daily at 0000 and 1200 UTC, followed by 10-day global numerical forecasts initiated from each assimilation time.

3.2 *Global forecast statistical analysis*

3.2.1 *O − B statistics*

Using 0000 UTC on 30 August 2022 as a representative case, Fig. 3 demonstrates the spatial distribution of observation-minus-background (O − B) differences for *MetOp-C* AMSU-A Channel 6 (54.400 GHz) simulations generated by both RTTOV and ARMS. Comparative analysis between Fig. 3a and Fig 3b reveals similarities in both the spatial distribution patterns and magnitude ranges of (O − B) values produced by the two radiative transfer models. Figure 3c presents the differential (O − B) results obtained by subtracting ARMS simulations from RTTOV outputs. The experimental results demonstrate that the systematic bias between RTTOV and ARMS simulations remains relatively small, with values predominantly ranging between −0.05 and 0.05 K. A notable concentration of brightness temperature biases is observed over land areas between 20°–40°N and 70°–120°E, where maximum biases approxim-

Table 2. Differences between ARMS and RTTOV11.2

Item	ARMS	RTTOV11.2
Training dataset	101 levels ECMWF 83 profiles	54 levels ECMWF 83 profiles
LBL model	MPM (Liebe, 1989)	AMSUTRAN (Turner et al., 2019)
Transmittance fitting method	Dry air: ODPS Water vapour: ODAS	ODPS (Chen et al., 2010)
Atmospheric profile interface	Layer temperature (K) Layer water vapour (g kg^{-1}) Layer ozone (ppmv)	Level temperature (K) Level water vapour (kg kg^{-1} or ppmv) Level ozone (ppmv)
Hydrometeors interface	Cloud water, cloud ice, rain, snow, graupel, and hail	Cloud water, cloud ice, rain, and snow
Radiative transfer solver	P2S, ADA, VDA, VDISORT	delta-Eddington
Sea surface emissivity	FASTEM6	FASTEM5

Note: MPM: Millimeter-wave Propagation Model; ODPS: Optical Depth in Pressure Space; ODAS: Optical Depth in Absorber Space; and P2S: Polarimetric Two-Stream.

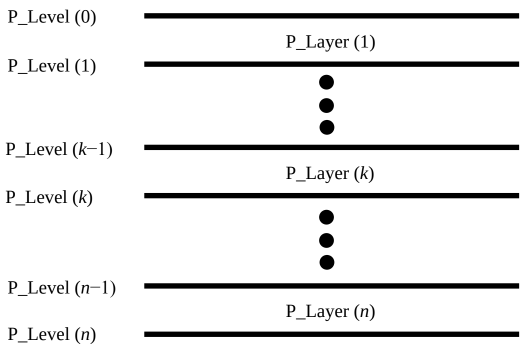

Fig. 2. Schematic diagram of atmospheric vertical discretization, illustrating the pressure level (P_Level) and pressure layer (P_Layer) configurations.

Table 3. Experimental design

Experiment name	Observation data	RT model
Test_arms	Conventional observations + MetOp-C AMSU-A	ARMS1.2
Test_rttov	Conventional observations + MetOp-C AMSU-A	RTTOV11.2

ately 0.05 K. This spatial pattern of discrepancies may be attributed to several factors: (1) the presence of cloud cover and precipitation within this region, which introduce complexities in radiative transfer calculations; (2) potential differences in the treatment of cloud–radiation interactions between the two models; and (3) variations in the parameterization of atmospheric absorption and scattering properties under cloudy conditions. Over oceanic regions near 60°S, both models exhibit brightness temperature biases reaching 0.05 K. This phenomenon may be associated with the combined effects of cold sea surface temperatures, which can influence surface emissivity calculations and subsequent radiative transfer simulations. The complex interplay between sea surface properties and lower atmospheric conditions in this region may contribute to the observed discrepancies between the two models' simulations.

Within the tropical region (30°S–30°N), satellite observations near the swath edges exhibit systematic negative biases of approximately −0.05 K, primarily attributable to larger zenith angles. The calculation of sea surface emissivity through the FAST Emissivity Model (FASTEM) model requires precise inputs of both satellite scan geometry and sea surface parameters. Consequently, increased scan angles at swath edges can affect the accuracy of surface emissivity parameterizations, thereby influencing the radiative transfer solver's performance in accurately computing radiative transfer equations. Furthermore, in the western Pacific region, the presence of Typhoon Hinnamnor introduces additional complexities. The cloud structures and precipitation systems associated with the typhoon create substantial uncertainties in the background field, potentially leading to biased temperature and humidity profiles within the ty-

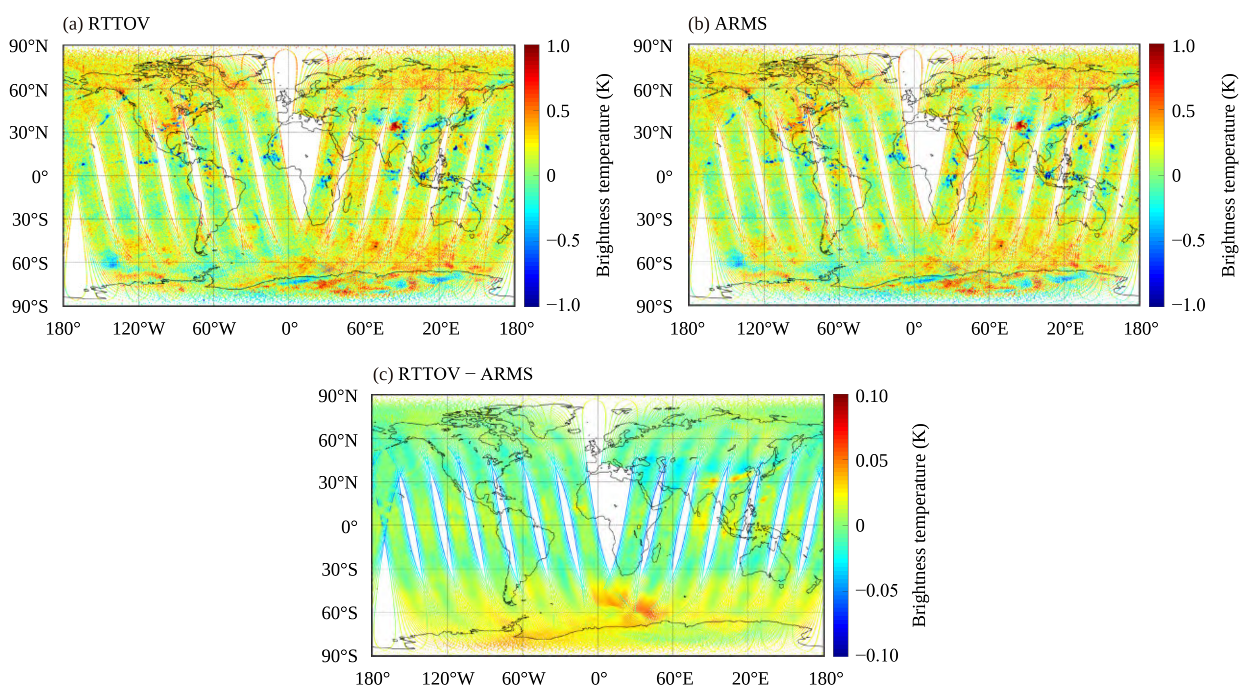

Fig. 3. Spatial distributions of observation-minus-background (O − B) brightness temperature (BT, unit: K) for *MetOp-C* AMSU-A Channel 6 (54.400 GHz) simulations at 0000 UTC 30 August 2022 from (a) the RTTOV model and (b) the ARMS model. (c) Spatial distribution of the difference of (O − B) BT between RTTOV and ARMS. All simulations were generated by using the same background field.

phoon-affected area. These atmospheric conditions contribute to discrepancies in AMSU-A brightness temperature simulations between ARMS and RTTOV. Despite these localized biases in specific regions, including high-latitude areas and typhoon-affected zones, the overall consistency between ARMS and RTTOV (O − B) simulation results demonstrates that ARMS achieves comparable satellite brightness temperature simulation capabilities to the established RTTOV model. This performance parity is particularly evident in the global domain, where both models exhibit similar simulation accuracy.

To provide a comprehensive evaluation of the differences between ARMS and RTTOV, Fig. 4 presents the statistical distribution of (O − B) values, including both means and standard deviations, under clear-sky conditions during the assimilation of *MetOp-C* AMSU-A radiance data. It should be noted that four microwave window channels: Channel 1 (23.800 GHz), Channel 2 (31.400 GHz), Channel 3 (50.300 GHz), and Channel 15 (89.000 GHz) are specifically designated for cloud and precipitation detection purposes. As these channels are excluded from the assimilation process, they have been consequently omitted from the statistical analysis presented in this study.

Figure 4a demonstrates that over oceanic regions, both ARMS and RTTOV exhibit minimal (O − B) biases across most channels. Specifically, Channels 4 through 13 show mean (O − B) biases below 0.1 K, while Channel 14 displays a slightly elevated bias of −0.2 K. This discrepancy can be primarily attributed to the implementation of different versions of the fast emissivity model, with RTTOV employing FASTEM5 and ARMS utilizing the more recent FASTEM6. Under clear-sky conditions, the accuracy of sea surface microwave radiation simulations is predominantly governed by surface emissivity calculations. In contrast, over land surfaces (Fig. 4b), Channels 5 to 14 maintain mean (O − B) biases within 0.1 K, demonstrating robust performance for both models. However, Channel 4 exhibits significantly larger biases, with ARMS reaching approximately 0.3 K and RTTOV slightly higher at 0.35 K. This enhanced sensitivity stems from Channel 4's central frequency of 52.800 GHz, which peaks in energy contribution near the surface at approximately 1000 hPa. As a near-surface channel, its strong dependence on surface emissivity properties inherently increases susceptibility to (O − B) biases, particularly over land surfaces where emissivity characterization presents greater challenges compared to

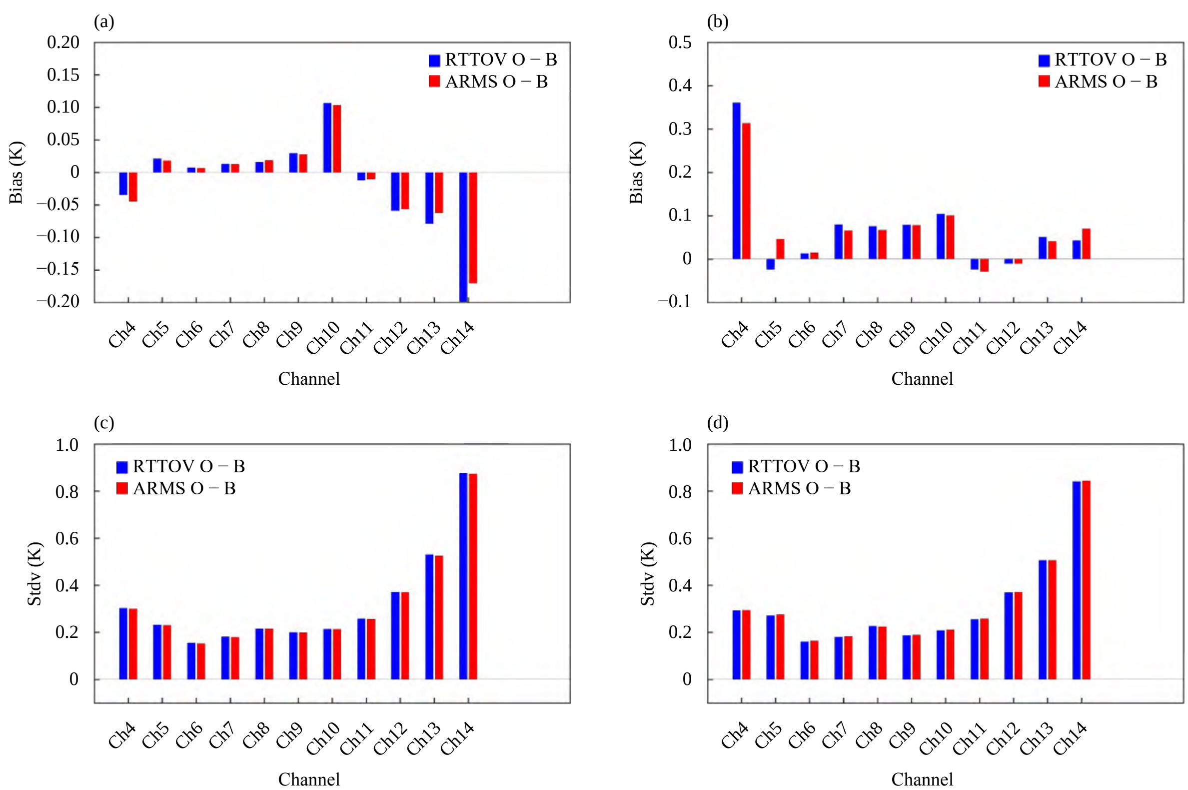

Fig. 4. Statistical analysis of (O − B) brightness temperature (BT) values for *MetOp-C* AMSU-A radiance assimilation at 0000 UTC 30 August 2022 using both RTTOV (blue) and ARMS (red). (a, b) Variations of (O − B) BT bias with different channels over (a) oceanic and (b) land regions. (c, d) Variations of (O − B) BT standard deviation with channels over (c) oceanic and (d) land regions.

ocean surfaces.

Regarding the (O − B) standard deviations (Figs. 4c, d), both ARMS and RTTOV demonstrate remarkably consistent statistical characteristics across land and oceanic surfaces. The (O − B) standard deviations for Channels 4 through 13 remain below 0.5 K for both models, indicating stable performance across these atmospheric sounding channels. However, Channel 14 exhibits a comparatively higher standard deviation of approximately 0.8 K, which can be attributed to its spectral characteristics. Operating at a central frequency of 57.290 GHz with a narrow bandwidth of 0.322 ± 0.004 GHz, Channel 14's inherent technical specifications result in increased observation noise levels. This enhanced noise susceptibility represents a fundamental limitation in the channel's observational precision, directly impacting the simulation accuracy of both radiative transfer models.

To quantitatively assess the performance of ARMS within the YH4DVAR system, this study analyzed the statistical characteristics of (O − B) values for *MetOp-C* AMSU-A Channels 5, 6, and 7 during the period from 1 to 30 August 2022, as presented in Fig. 5. The analysis of Figs. 5a–c demonstrates that the ARMS produced (O − B) mean biases of 0.01 K, 0.02 K, and 0.03 K for Channels 5, 6, and 7, respectively, while RTTOV yielded corresponding values of 0.01 K, 0.01 K, and 0.02 K. These results indicate statistically equivalent performance between the two models in terms of mean bias. It is noteworthy that during the initial testing period (1–2 August), both RTTOV and ARMS exhibited a gradual decrease in their (O − B) mean values, which stabilized after 3 August and remained consistently near 0 K throughout the subsequent testing period. This stabilization pattern can be attributed to the initialization of variational bias correction coefficients for both RTTOV and ARMS on 1 August. During the initial assimilation cycles, this initialization resulted in relatively higher (O − B) biases. However, as the assimilation system progressed, continuous adjustments and updates to the bias correction coefficients effectively reduced and eventually stabilized the (O − B) mean biases. Regarding the (O − B) standard deviation (Figs. 5d–f), ARMS demonstrated identical performance to RTTOV across all three channels, with values of 0.25 K, 0.16 K, and 0.19 K for Channels 5, 6, and 7, respectively. This consistency in standard deviations further confirms the comparable performance of ARMS to that of RTTOV.

3.2.2 *Results of variational bias correction*

Variational data assimilation assumes that the errors in the model background and observations are unbiased and follow a Gaussian distribution. However, in reality, systematic biases exist between satellite-observed brightness temperatures and those simulated by radiative transfer models and are related to instrumental errors, errors in forwards models, NWP background errors and uncertainties in clouds and precipitation. These systematic biases need to be corrected before assimilation to meet the theoretical assumptions of variational data assimilation. In the YH4DVAR system, variational bias correction (Auligné et al., 2007) is employed to adjust for such systematic errors.

Using *MetOp-C* AMSU-A Channel 6 observations

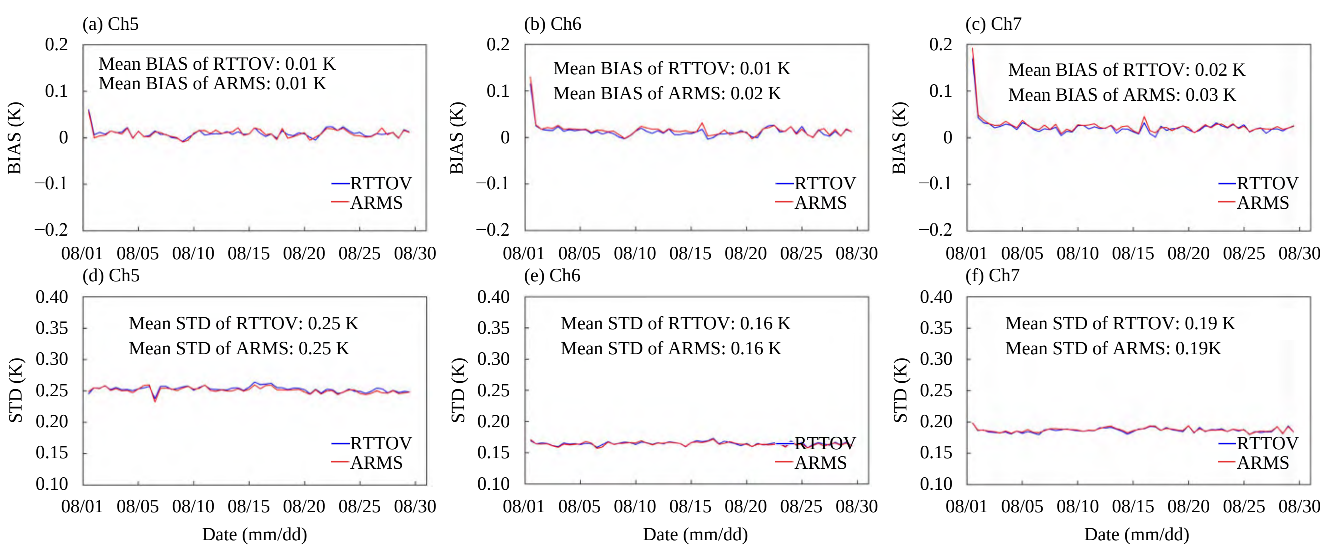

Fig. 5. Statistical analysis of *MetOp-C* AMSU-A radiance (unit: K) data for selected channels during 1–30 August 2022. (a–c) Temporal evolution of (O − B) mean values (BIAS) for Channel 5 (53.596 ± 0.115 GHz), Channel 6 (54.400 GHz), and Channel 7 (54.940 GHz). (d–f) Temporal evolution of (O − B) standard deviations (STD) for the corresponding channels.

from 0000 UTC on 30 August 2022 as a representative case, this study examines the performance of variational bias correction (VarBC) within the YH4DVAR system for both RTTOV and ARMS, as illustrated in Fig. 6. Prior to bias correction, RTTOV exhibited (O − B) statistics with a mean of 1.39 K, standard deviation of 0.32 K, and root mean square error (RMSE) of 1.42 K. Following VarBC implementation, these metrics significantly improved to 0.13 K (mean), 0.16 K (standard deviation), and 0.16 K (RMSE). ARMS demonstrated comparable performance under the same VarBC framework, achieving post-correction (O − B) statistics of 0.13 K (mean), 0.16 K (standard deviation), and 0.16 K (RMSE). Figures 6e (RTTOV) and 6f (ARMS) display the probability density function (PDF) distributions of Channel 6 (O − B) , with blue representing prebias correction and red indicating postbias correction PDFs. The (O − B) data from both the RTTOV and the ARMS clearly show unbiased Gaussian distribution characteristics after bias correction, indicating an ideal correction outcome. Table 4 summarizes the (O − B) mean, standard deviation, and RMSE for AMSU-A Channels 4–14 before and after bias correction. Overall, the ARMS adapts well to the variational bias correction technique within the YH4DVAR system, effectively completing the bias correction process.

3.2.3 *Results of quality control*

Owing to the influence of observation errors, topography, complex underlying surfaces, clouds, and precipitation, satellite microwave observation data often contain significant inaccuracies. It is necessary to employ a quality control scheme to exclude anomalous data. The

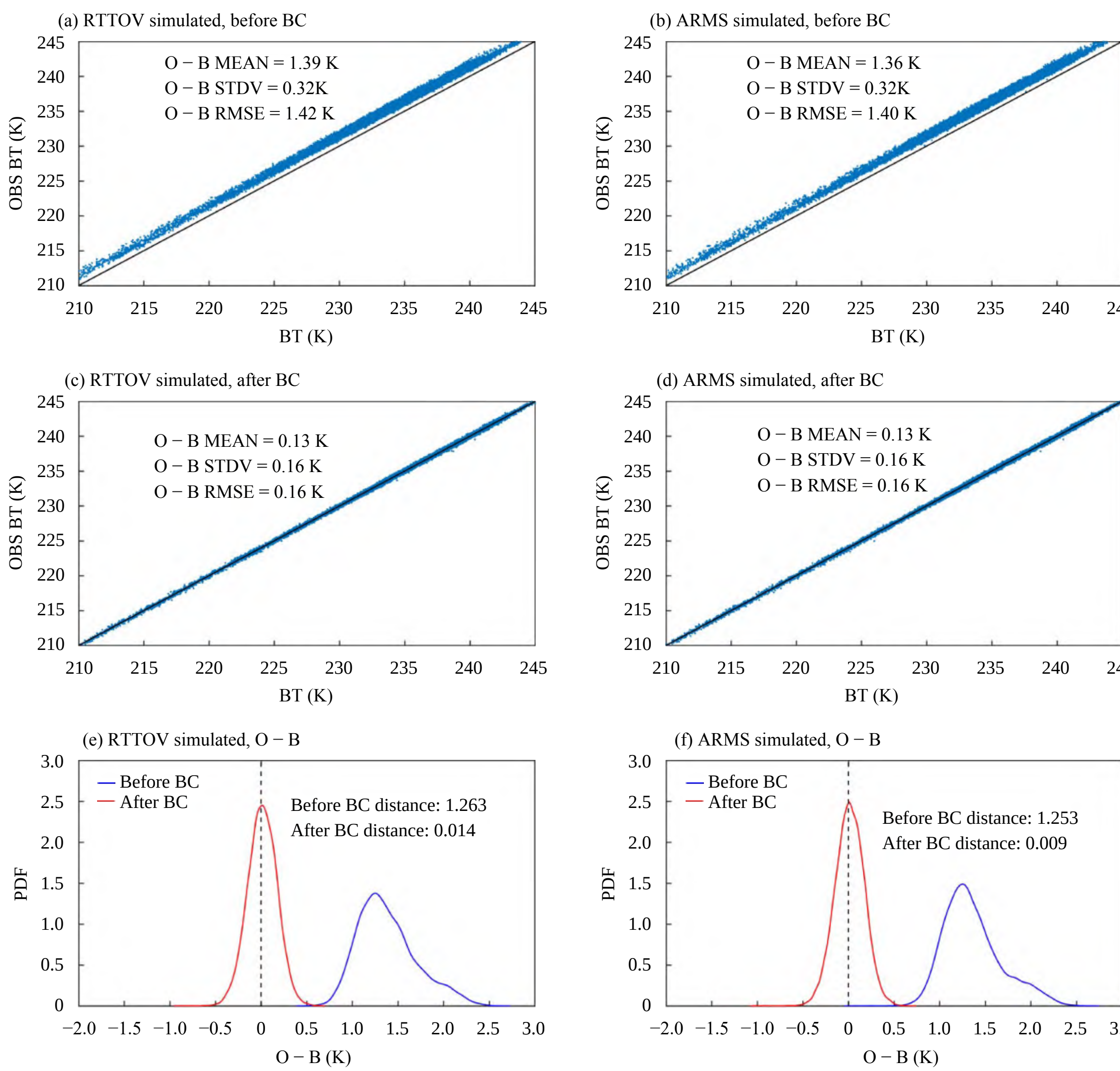

Fig. 6. Scatter plots of observed vs. simulated brightness temperature (BT, unit: K) from (a, c) RTTOV and (b, d) ARMS for the *MetOp-C* AMSU-A Channel 6 (54.400 GHz) (a, b) before and (c, d) after variational bias correction (BC). (e, f) The associated probability density function (PDF) of (O – B) BT values before (blue) and after (red) BC for (e) RTTOV and (f) ARMS.

Table 4. Means (MEAN), standard deviations (STD), and root mean square errors (RMSE) of (O − B) brightness temperature (unit: K) for *MetOp-C* AMSU-A Channels 4–14 on 30 August 2022, before and after bias correction

Channel	Frequency (GHz)	RTTOV O − B (MEAN, STD, RMSE)		ARMS O − B (MEAN, STD, RMSE)	
		Before BC (K)	After BC (K)	Before BC (K)	After BC (K)
Ch4	52.800	(0.81, 0.64, 0.98)	(0.26, 0.31, 0.31)	(0.89, 0.67, 1.07)	(0.26, 0.31, 0.31)
Ch5	53.596 ± 0.115	(0.85, 0.49, 0.98)	(0.20, 0.25, 0.25)	(0.85, 0.47, 0.96)	(0.20, 0.24, 0.25)
Ch6	54.400	(1.39, 0.32, 1.42)	(0.13, 0.16, 0.16)	(1.36, 0.32, 1.40)	(0.13, 0.16, 0.16)
Ch7	54.940	(1.59, 0.32, 1.62)	(0.15, 0.18, 0.19)	(1.57, 0.32, 1.60)	(0.15, 0.18, 0.19)
Ch8	55.500	(0.80, 0.37, 0.88)	(0.18, 0.22, 0.21)	(0.81, 0.37, 0.89)	(0.18, 0.22, 0.22)
Ch9	57.290 (f0)	(0.86, 0.29, 0.91)	(0.16, 0.19, 0.20)	(0.86, 0.30, 0.91)	(0.16, 0.19, 0.20)
Ch10	f0 ± 0.217	(0.51, 0.32, 0.60)	(0.19, 0.22, 0.24)	(0.53, 0.31, 0.61)	(0.18, 0.22, 0.23)
Ch11	f0 ± 0.322 ± 0.048	(0.39, 0.35, 0.48)	(0.21, 0.26, 0.25)	(0.41, 0.35, 0.50)	(0.21, 0.26, 0.26)
Ch12	f0 ± 0.322 ± 0.022	(0.67, 0.45, 0.79)	(0.30, 0.38, 0.36)	(0.68, 0.46, 0.79)	(0.30, 0.38, 0.38)
Ch13	f0 ± 0.322 ± 0.010	(0.95, 0.61, 1.10)	(0.44, 0.55, 0.56)	(1.02, 0.61, 1.17)	(0.43, 0.55, 0.55)
Ch14	f0 ± 0.322 ± 0.004	(0.93, 1.24, 1.46)	(0.74, 0.94, 0.95)	(0.99, 1.13, 1.34)	(0.73, 0.93, 0.94)

quality control procedure for AMSU-A in YH4DVAR is as follows. (1) Channel 4 to Channel 5 observational data beyond 60° latitude in both the southern and northern hemispheres are removed. (2) Channel 4 to Channel 5 observations over coastlines and sea ice surfaces are removed. (3) The first three observational pixels at the edge of the scan line are excluded. (4) Channel 5 is not used if the topography exceeds 500 m, whereas Channel 6 is avoided if the elevation exceeds 1500 m. (5) For Channel 4, if the absolute value of (O − B) is greater than 0.7 K over oceans or greater than 1.2 K over lands, data from Channels 5–7 are discarded. (6) To eliminate the impacts of clouds and precipitation on the AMSU-A data, observations where the cloud liquid water path (CLWP) value exceeds 0.03 kg m^{-2} over oceanic areas are excluded. Over lands, the observations are screened by using the scattering index (SI), and those with an SI exceeding 30 are excluded from the AMSU-A data. (7) Data are removed from window channels and Channels 1–3 and 15.

Using the assimilation cycle at 0000 UTC on 30 August 2022 as a representative case, Fig. 7 illustrates the distribution of quality-controlled data in the YH4DVAR assimilation system, with Figs. 7a and 7b specifically showing the results over land and oceanic regions, respectively. The analysis reveals a distinct spatial pattern, with significantly higher data retention over oceanic areas compared to land surfaces. This disparity primarily stems from the fundamental differences in surface characteristics: oceanic surfaces exhibit relatively stable radiative properties that can be accurately simulated using FASTEM microwave emissivity models, thereby allowing a greater proportion of satellite observations to pass quality control thresholds. In contrast, land surfaces present substantial challenges due to their inherent complexity and heterogeneity. The diverse composition of vegetation types, soil properties, and urban structures, combined with the dynamic nature of land surface parameters (including soil type, surface temperature, and moisture content), creates significant variability in microwave land surface emissivity. This variability often leads to (O − B) anomalies that exceed quality control thresholds, resulting in the exclusion of most land surface observations from the assimilation system.

Figure 8 illustrates the temporal variation in assimilated data counts for *MetOp-C* AMSU-A Channels 5, 6, and 7 within the YH4DVAR system during the period from 1 to 30 August 2022. When ARMS was implemen-

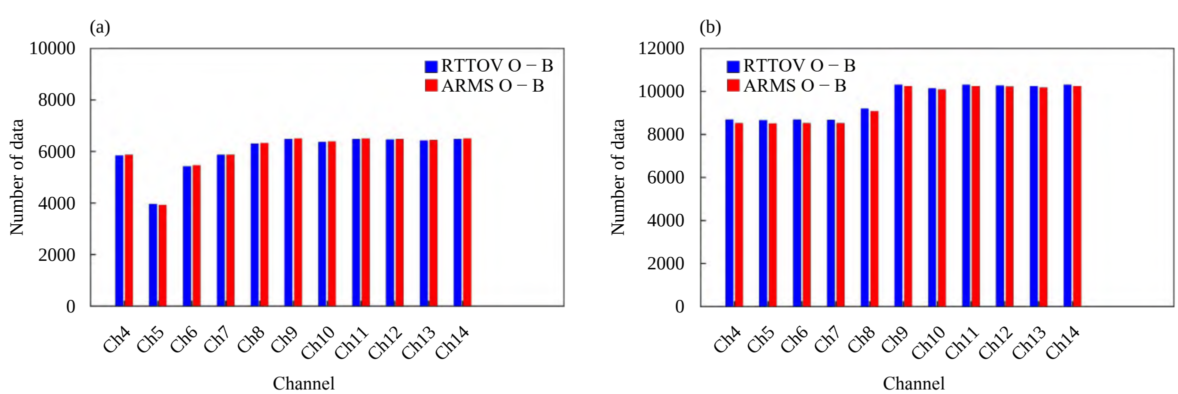

Fig. 7. Distribution of assimilated *MetOp-C* AMSU-A observation counts over channels at 0000 UTC 30 August 2022 over (a) land and (b) ocean.

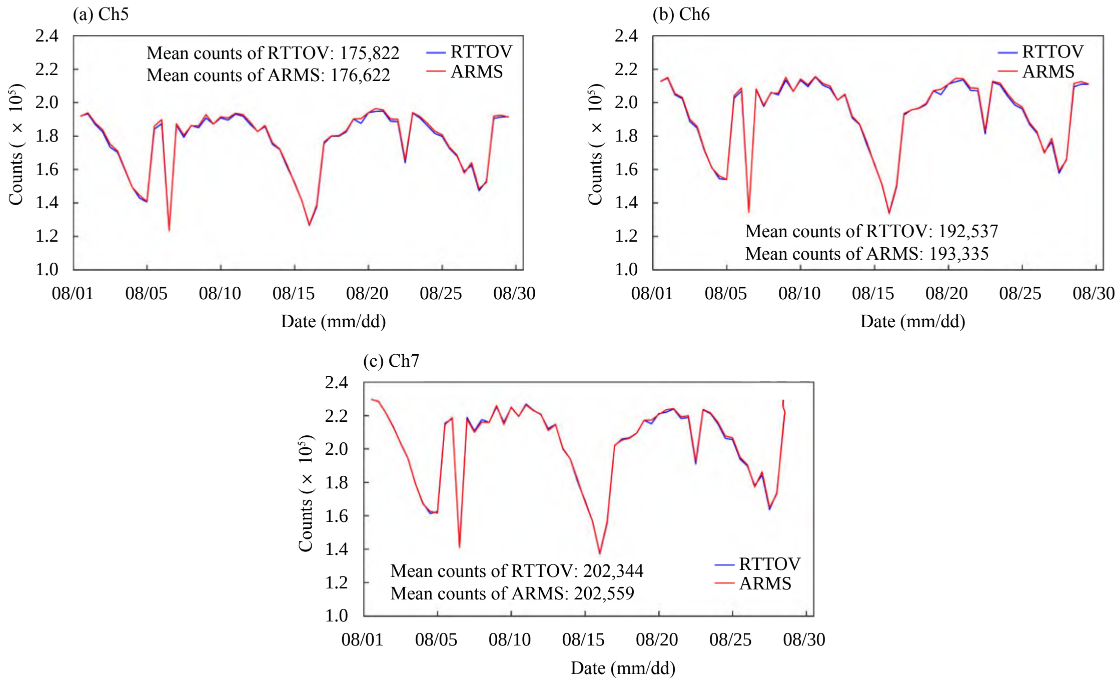

Fig. 8. Temporal variations of quality-controlled *MetOp-C* AMSU-A observation counts assimilated into YH4DVAR (RTTOV: blue; ARMS: red) during 1–30 August 2022 for (a) Channel 5 (53.596 ± 0.115 GHz), (b) Channel 6 (54.400 GHz), and (c) Channel 7 (54.940 GHz).

ted as the observation operator, the system processed daily averages of 176,622, 193,335, and 202,559 observations for Channels 5, 6, and 7, respectively. Comparatively, RTTOV processing yielded corresponding daily averages of 175,822, 192,537, and 202,344 observations for the same channels. Statistical analysis demonstrates nearly identical data quality control outcomes between the two observation operators, with a minimal discrepancy of less than 0.5% in processed data volume throughout the one-month evaluation period.

3.2.4 *Results of assimilation and forecasting*

In this paper, we conducted a comprehensive evaluation of AMSU-A data assimilation performance using both ARMS and RTTOV observation operators across three key regions: the Northern Hemisphere (NthH), Southern Hemisphere (SthH), and the tropical zone (TrpC) during the period from 1 to 31 August 2022. The resulting analysis fields were rigorously validated against the ECMWF Reanalysis version 5 (ERA5) dataset, which served as the reference truth. To quantify the analysis accuracy, RMSE was computed for four critical meteorological variables: geopotential height, temperature, and horizontal wind components (*U* and *V*). The comparative results are presented in Fig. 9, with (a) geopotential height, (b) temperature, (c) *U*-wind, and (d) *V*-wind.

The analysis of geopotential height assimilation results (Fig. 9a) demonstrates that both ARMS and RTTOV produce analysis fields with comparable RMSE values across all pressure levels, indicating statistically equivalent performance in geopotential height field assimilation. Consistent results were observed for the temperature fields and the *U* and *V* wind fields. These comprehensive comparisons reveal that ARMS achieves assimilation quality comparable to the established RTTOV model across all evaluated regions, including the Northern Hemisphere, Southern Hemisphere, and tropical zone, demonstrating its robustness as a radiative transfer model for global data assimilation applications.

After assimilation process, this study employed the YHGSM to generate 240-h numerical forecasts from the analysis fields. All forecast results from August 2022 were validated against the ERA5 reanalysis dataset. Figure 10 displays the temporal evolution of the anomaly correlation coefficient (ACC) for 500-hPa geopotential height forecasts across three domains: the Northern Hemisphere, Southern Hemisphere, and Tropical zone. The forecast skill comparison between RTTOV (blue line) and ARMS (red line) reveals the following key findings. (1) In both the Northern and Southern Hemispheres, ARMS demonstrates forecast skill equivalent to RTTOV; and (2) in tropical zone, ARMS and RTTOV exhibit comparable ACC values within the initial 72-h forecast window, beyond which ARMS maintains a marginally higher ACC than RTTOV.

Figure 11 shows the forecast scorecard when the ARMS is employed as the observational operator in YH4DVAR for the Northern Hemisphere, Southern

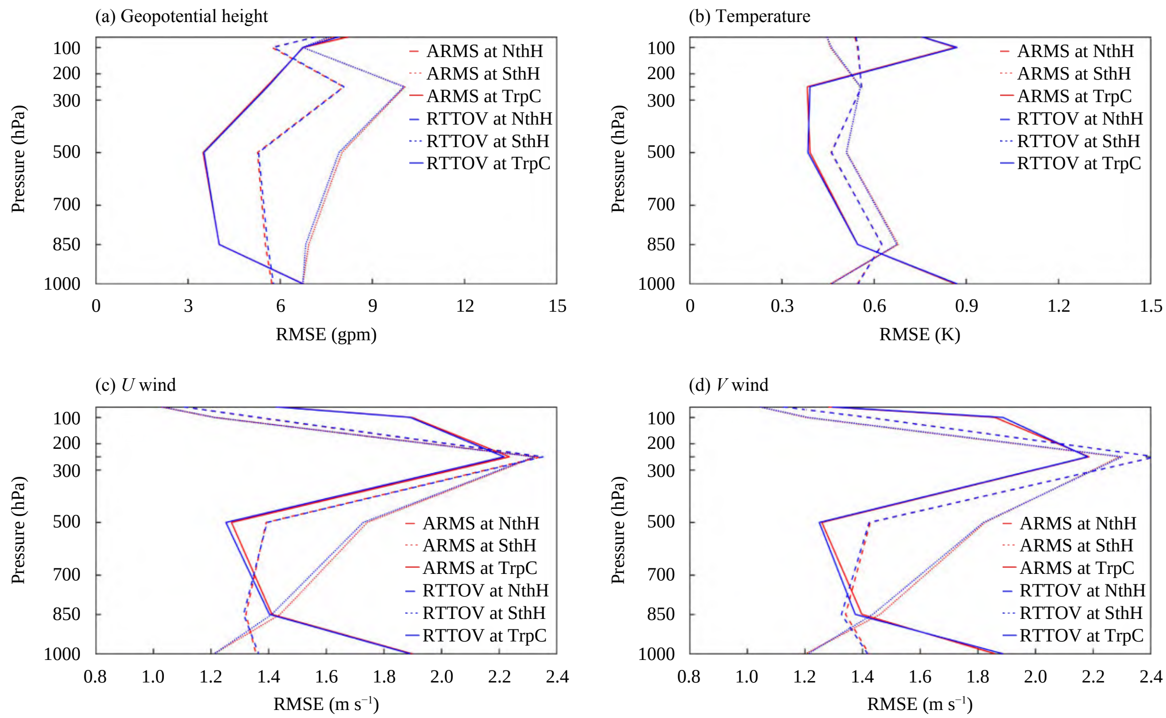

Fig. 9. Vertical profiles of RMSEs for the analysis fields of (a) geopotential height, (b) temperature, (c) *U* wind, and (d) *V* wind, based on the ARMS (red) and RTTOV (blue) assimilation results averaged over 1–30 August 2022.

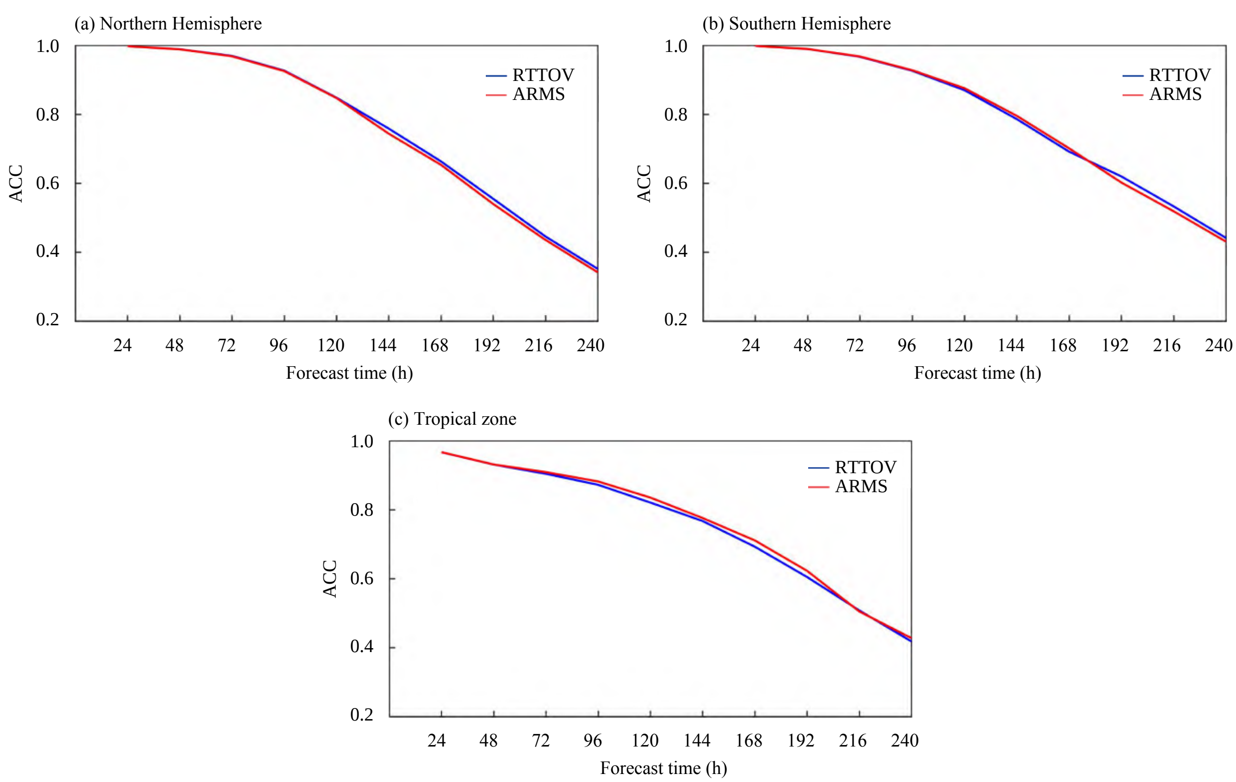

Fig. 10. Temporal evolution of anomaly correlation coefficient (ACC) for 500-hPa geopotential height forecasts using RTTOV (blue) and ARMS (red) during 1–30 August 2022 over the (a) Northern Hemisphere, (b) Southern Hemisphere, and (c) tropical zone.

Hemisphere, and tropical zone. The scorecard reveals that ARMS delivers forecast quality comparable to RTTOV in most regions, with particularly improvements in tropical zone predictions for the 850-hPa tem-

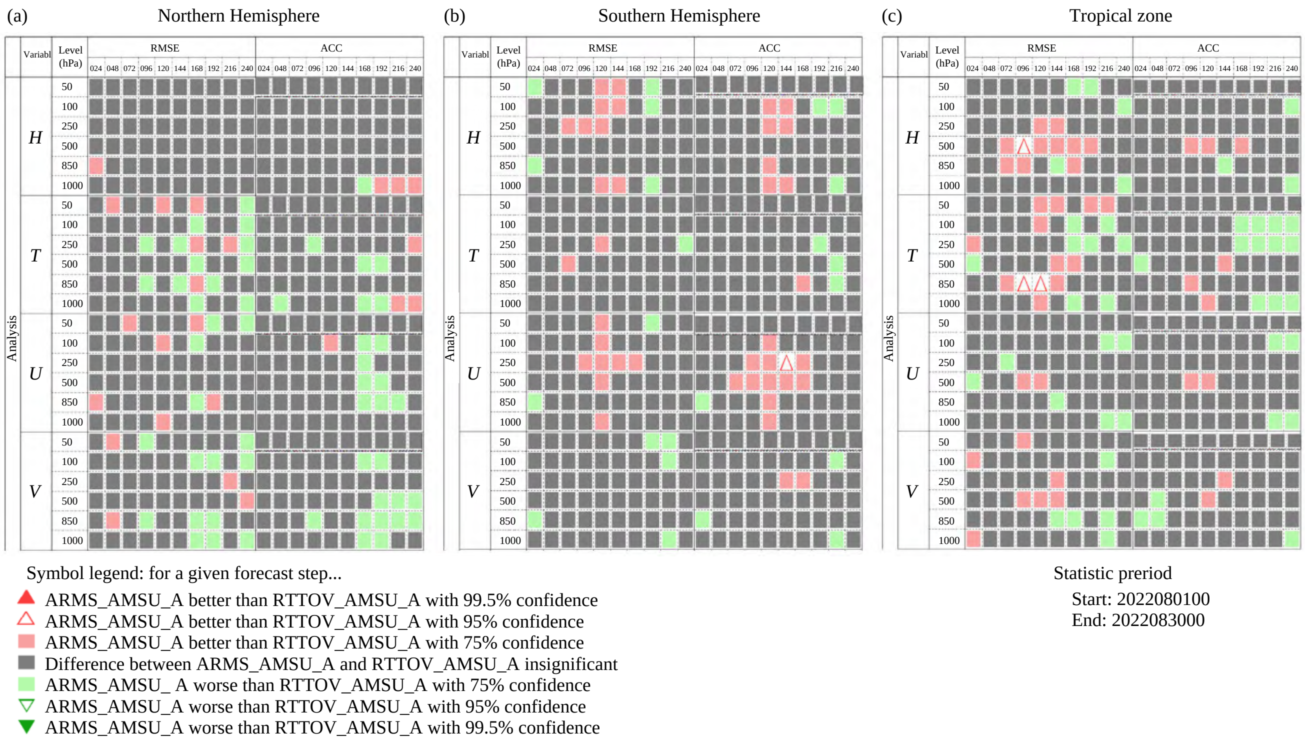

Fig. 11. Forecast scorecard showing a comparison between ARMS and RTTOV during 1–30 August 2022 for the (a) Northern Hemisphere, (b) Southern Hemisphere, and (c) tropical zone.

perature field and 500-hPa geopotential height field. These enhancements are primarily driven by ARMS's implementation of the advanced FASTEM6 sea surface emissivity model, which offers superior accuracy in radiative transfer calculations compared to the FASTEM5 model used in RTTOV11.2. The improved emissivity parameterization in ARMS enables more precise simulations of microwave radiation over oceanic surfaces, thereby enhancing forecast accuracy in tropical regions. In the Northern Hemisphere, ARMS demonstrates forecast performance equivalent to RTTOV, a result likely influenced by the region's significant landmass coverage. This similarity stems from YH4DVAR's implementation of the dynamic inversion method for land surface emissivity calculation by Karbou et al. (2005), which ensures consistent emissivity inputs for both ARMS and RTTOV over terrestrial surfaces. The forecast scorecard thus demonstrates ARMS's capability to match RTTOV's performance across diverse regions while offering potential improvements in tropical zone predictions through its advanced radiative transfer modeling framework.

3.2.5 *CPU calculation time statistics*

Computational efficiency represents a crucial performance metric for evaluating data assimilation systems. To assess this aspect, we conducted comprehensive tests on a high-performance computing cluster featuring Intel Xeon Gold 6150 processors (experimental configuration detailed in Table 5) , comparing the computational requirements of YH4DVAR when utilizing RTTOV versus ARMS as observation operators during the period from 1 to 30 August 2022. The analysis reveals that ARMS demonstrates computational efficiency comparable to RTTOV, with average computation times of 167 minutes and 166 minutes, respectively. However, both systems exhibited peak computation times of approximately 190 minutes during specific assimilation cycles at 0000 UTC on 20 August and 29 August. These performance anomalies are likely attributable to computational node instabilities within the server infrastructure rather than inherent differences in the observation operators' computational requirements. The overall computational performance comparison, as illustrated in Fig. 12, confirms that ARMS achieves operational efficiency equivalent to the established RTTOV model within the YH4DVAR framework, demonstrating its suitability for operational numerical weather prediction systems.

Table 5. Operational configuration of the CPU calculation

CPU	Base frequency	Number of nodes	Processes per node	Omp threads	Total cores	Horizontal resolution
Intel™ Xeon™ Gold 6150	2.7 GHz	8	16	2	256	T799 (approximately 25 km)

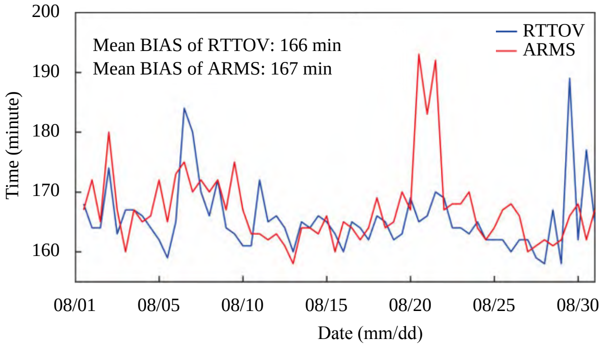

Fig. 12. Temporal evolution of computational time consumed by the YH4DVAR system using RTTOV (blue) and ARMS (red) observation operators during 1–30 August 2022.

3.3 *Typhoon case study*

This paper employs Typhoon Hinnamnor (the 11th typhoon of 2022) as a case study to evaluate the performance of the ARMS within the YH4DVAR system for *MetOp-C* AMSU-A data assimilation. Typhoon Hinnamnor initially formed at 0600 UTC on 28 August 2022 and underwent gradual intensification in subsequent days, reaching super typhoon status as defined by the CMA by 1800 UTC on 30 August. During its development phase, the typhoon's trajectory was significantly influenced by a tropical depression located to its southwest (approximately 20°N, 130°E), resulting in a characteristic southwestward turn due to the Fujiwhara effect. This interaction culminated in the absorption of the tropical depression by Hinnamnor on 31 August. Subsequently, the typhoon's trajectory exhibited a notable northward turn around 0000 UTC on 2 September, forming a distinct "V-shaped" trajectory under the combined influence of a midlatitude westerly trough and the subtropical high.

3.3.1 *Results of analysis innovation*

At 0000 UTC on 30 August 2022, Typhoon Hinnamnor (centered at 26.8°N, 137.4°E) approached a tropical depression system located at 20°N, 130°E. This paper examines the spatial distribution of quality-controlled *MetOp-C* AMSU-A Channel 6 observations assimilated through both RTTOV and ARMS within the YH4DVAR system. The analyzed data are superimposed on the Advanced Very High Resolution Radiometer (AVHRR) 12-μm spectral radiance imagery (shown as grey background in Fig. 13), which provides cloud distribution information for Typhoon Hinnamnor. In the deep convective region of the typhoon (approximately 15°–30°N, 130°–140°E), intense precipitation processes generate significant microwave radiation scattering, potentially contaminating microwave observation data. During clear-sky assimilation, rigorous quality control procedures are implemented to exclude precipitation-affected observations. The experimental results demonstrate that ARMS effectively identifies and filters out precipitation-contaminated data within typhoon-related cloud systems, matching RTTOV's performance in this critical aspect. Particularly noteworthy is both models' capability to retain valid observations from peripheral cloud systems distant from the typhoon center. The spa-

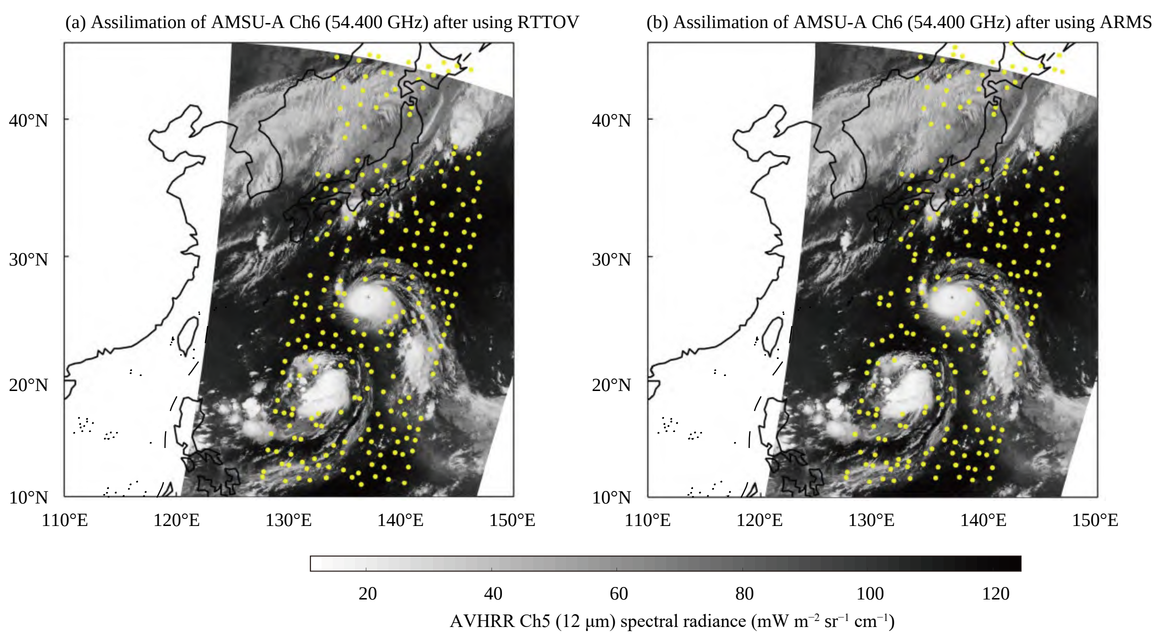

Fig. 13. Horizontal distributions of quality-controlled *MetOp-C* AMSU-A Channel 6 observations assimilated at 0000 UTC on 30 August 2022 (yellow dots) using (a) RTTOV and (b) ARMS. The background denotes the AVHRR 12-μm spectral radiance.

tial distribution patterns and data retention rates of AMSU-A observations in the Typhoon Hinnamnor-affected area show remarkable consistency between ARMS and RTTOV implementations. These findings indicate that ARMS achieves comparable performance to RTTOV in two key aspects: (1) simulating satellite-observed radiances under complex meteorological conditions; and (2) effectively filtering out erroneous observations while retaining high-quality data for assimilation.

The assimilation of satellite microwave temperature sounding data significantly enhances the model background field, particularly through improved representation of meteorological variables including geopotential height, temperature, and humidity fields. These enhancements facilitate more accurate simulations of atmospheric circulation patterns, ultimately improving typhoon forecast accuracy. Figure 14 presents the 500-hPa geopotential height analysis fields and their corresponding innovation at 0000 UTC on 30 August 2022. Comparative analysis of the geopotential height fields generated by RTTOV and ARMS assimilation within the YH4DVAR system reveals substantial consistency in spatial patterns. Both models capture the characteristic negative geopotential height increments associated with Typhoon Hinnamnor's central region (26.8°N, 137.4°E). Notably, the ARMS analysis field of geopotential height approximately 2 gpm lower than RTTOV at the typhoon center. This enhanced sensitivity in the ARMS analysis field suggests improved representation of the typhoon's dynamical structure, potentially leading to more accurate initialization of typhoon intensity in subsequent forecasts.

Figure 15 presents the analysis fields and corresponding increments of the 500-hPa temperature (upper panels) and humidity fields (lower panels) following assimilation. Both fields exhibit complex spatial distribution patterns, which can be attributed to three primary factors. (1) Radiative transfer model differences. RTTOV and ARMS employ distinct physical parameterizations and technical implementations, including differences in training datasets, numerical solution methods, predictor selection, and tangent-linear/adjoint operator formulations (see Table 1 for detailed comparisons). These fundamental differences in radiative transfer modeling approaches naturally lead to variations in the derived temperature and humidity fields. (2) Atmospheric thermodynamic interactions. The temperature and humidity fields are intrinsically coupled through complex physical processes. Temperature variations directly influence atmospheric saturation vapor pressure, thereby modulating relative humidity distributions. Conversely, humidity changes affect temperature fields through latent heat release/absorption during phase change processes. These interactions can amplify and propagate the initial differences arising from distinct radiative transfer models. (3) Nonlinear dynamics in typhoon systems. The highly nonlinear nature of typhoon dynamics can significantly amplify minor differences in (O − B) values (as shown in Fig. 13) throughout the assimilation process. This nonlinear amplification can lead to discernible differences in the analyzed geopotential height, temperature, and humidity fields. The experimental results demonstrate that the choice of radiative transfer model indeed influences the temperature and humidity analysis fields when assimilat-

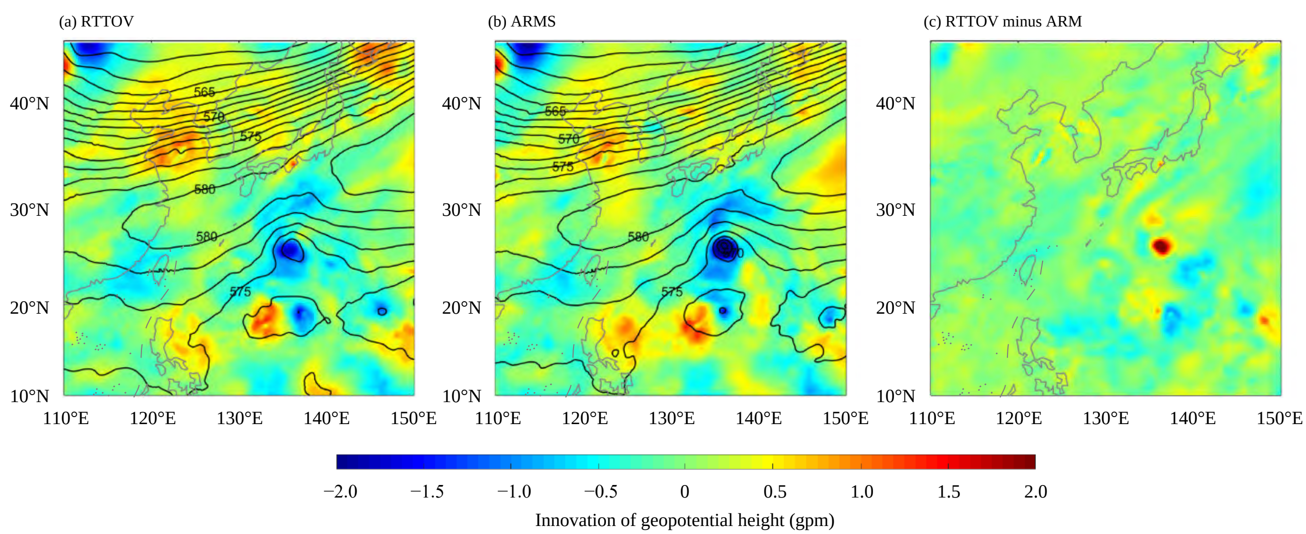

Fig. 14. (a, b) Distributions of 500-hPa geopotential height (contour, gpm) and its innovation (shaded, gpm) for assimilation using (a) RTTOV and (b) ARMS in the study domain for Typhoon Hinnamnor at 0000 UTC on 30 August 2022. (c) Difference of geopotential height innovation (shaded, gpm) between RTTOV and ARMS.

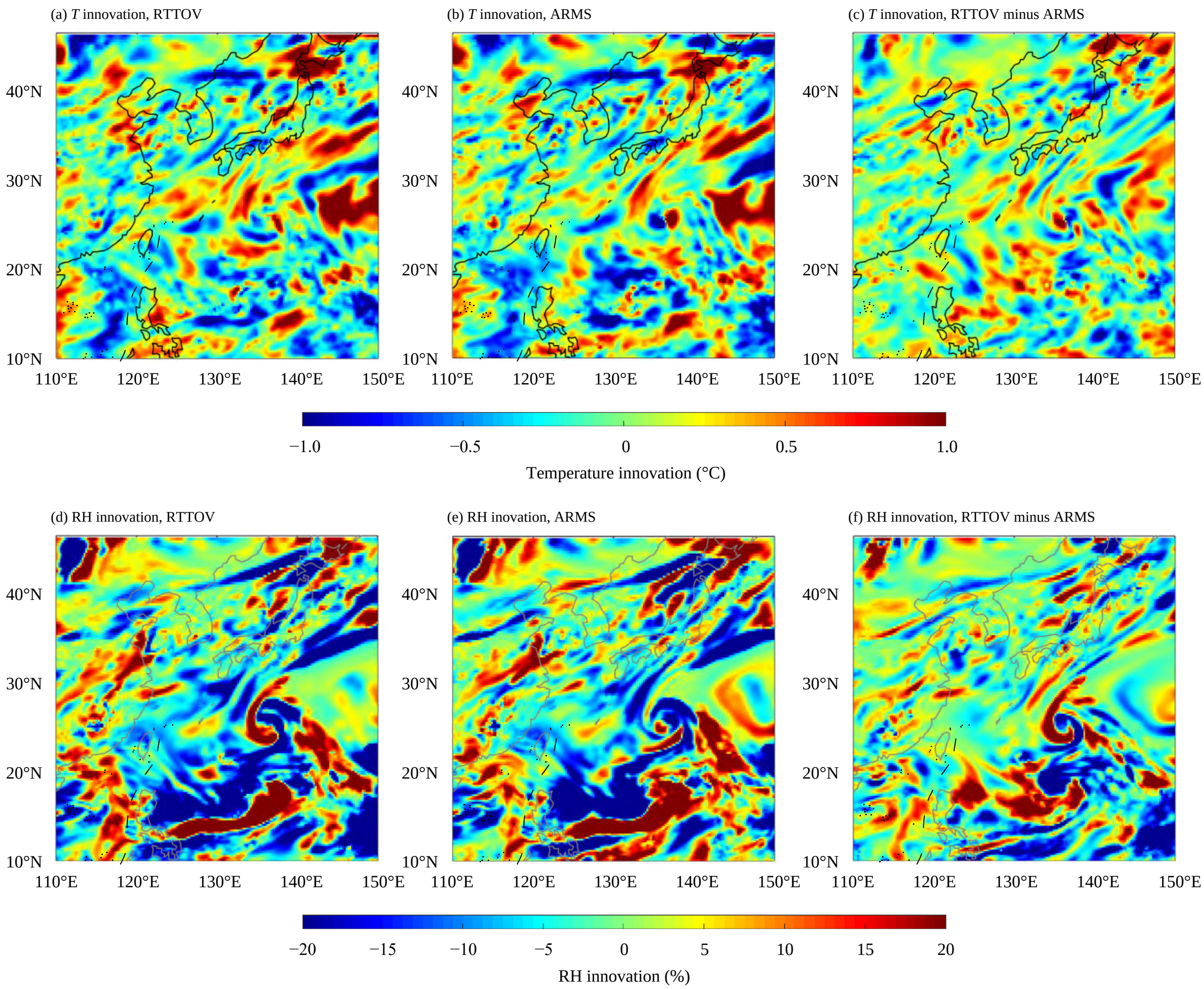

Fig. 15. Distributions of (a, b) temperature innovation (T, shaded, K) and (c, d) relative humidity innovation (RH, shaded, %) fields at 500 hPa, for assimilation using (a, d) RTTOV and (b, e) ARMS. (c, f) Corresponding differences between RTTOV and ARMS.

ing satellite microwave observations in typhoon-affected regions. However, the precise mechanisms and quantitative impacts of these differences warrant further investigation through targeted sensitivity studies and process-oriented diagnostics.

3.3.2 *Typhoon track and intensity errors*

During the period from 0000 UTC on 30 August to 1200 UTC on 31 August 2022, this paper conducted parallel assimilation experiments using AMSU-A data through two distinct observation operators, ARMS (represented by red trajectories) and RTTOV (blue trajectories) within the YH4DVAR system to evaluate their respective impacts on Typhoon Hinnamno's track forecasting. Figure 16 presents the comparative track forecasts generated by these two assimilation results. The analysis reveals remarkable consistency between ARMS and RTTOV in predicting the typhoon's trajectory, with both models successfully capturing the characteristic northward "V-shaped" occurring on 2 September. This consistent performance demonstrates that ARMS achieves typhoon track forecasting skill equivalent to the established RTTOV model, providing reliable operational guidance for tropical cyclone prediction.

Figure 17 presents a comprehensive statistical analysis of forecast errors for typhoon track, maximum wind speed, and minimum central pressure, with ARMS and RTTOV results depicted in red and blue, respectively. The initial position error at analysis time is approximately 60 km for both models (Fig. 17a). During the first 48 forecast hours, ARMS and RTTOV maintain similar track error characteristics, with mean errors around 60 km, though ARMS exhibits marginally lower errors. A significant error increase occurs between 48–72 h, coinciding with Hinnamnor's characteristic "V-shaped"

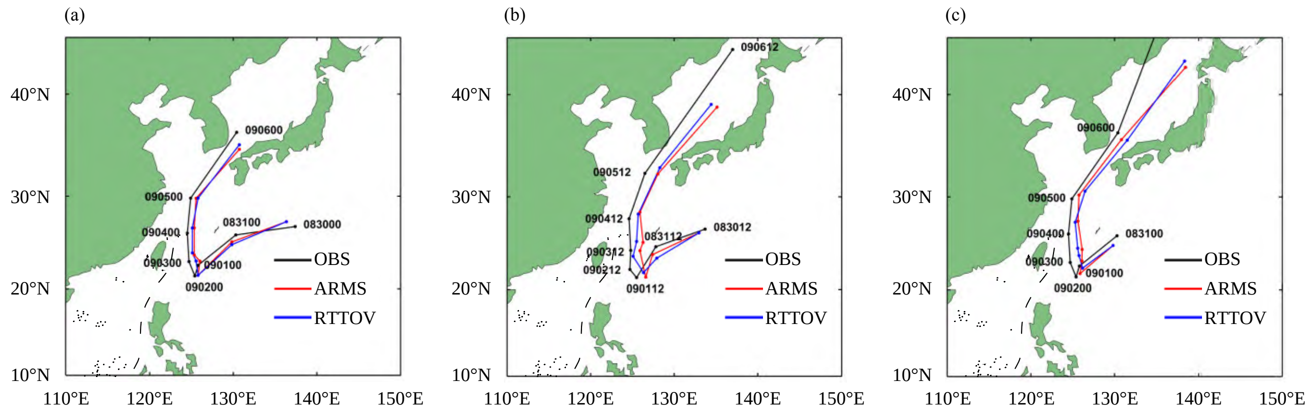

Fig. 16. Typhoon Hinnamor track predictions utilizing ARMS and RTTOV models at (a) 0000 UTC 30 August 2022; (b) 1200 UTC 30 August 2022; and (c) 0000 UTC 31 August 2022.

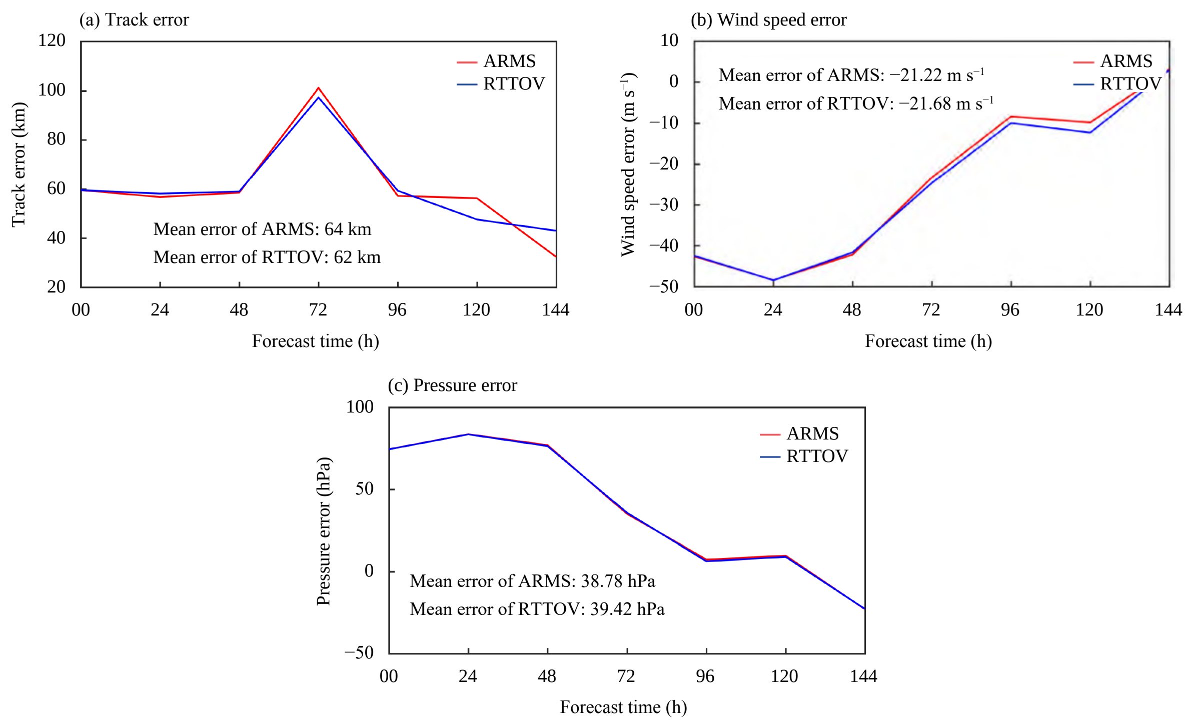

Fig. 17. Track and intensity forecast errors of Typhoon Hinnamnor. (a) Track error; (b) wind speed error; and (c) central pressure error.

northward turn on 2 September, when atmospheric circulation changes drive peak errors reaching 95 km for both models. As the typhoon subsequently follows a steady northward trajectory, track errors gradually decrease in both systems. The overall mean track errors are 64 km (ARMS) and 62 km (RTTOV), demonstrating statistically equivalent track forecasting performance between the two radiative transfer models.

Figures 17b and 17c present the comparative forecast error statistics for maximum wind speed and minimum central pressure between ARMS and RTTOV. The initial intensity forecast exhibits relatively large errors, with wind speed errors of 42 m s^{-1} and pressure deviations of approximately 76 hPa. Both models demonstrate decreasing intensity forecast errors with increasing lead time: maximum wind speed errors decrease to 2 m s^{-1} by 144 h, while minimum pressure errors reduce to approximately 2 hPa by 96–120 h.

Compared with RTTOV, the ARMS demonstrates certain improvements in typhoon intensity forecasting, with an average wind speed error reduction of approximately 2% and an average minimum pressure error reduction of approximately 1%. This improvement may be attributed to the FASTEM6 sea surface emissivity model used by the ARMS, which provides more accurate sea surface emissivity calculations than the FASTEM5 model used by RTTOV11.2 does, thereby slightly enhancing the typhoon intensity forecast.

4. Conclusions and discussion

This study implements the ARMS fast radiative transfer model within the YH4DVAR and conducts a comprehensive evaluation of its satellite data assimilation capabilities using *MetOp-C* AMSU-A observations. Through assessments of (O − B) spatiotemporal distributions, bias correction, quality control, innovation of assimilation, forecast skill, and computational efficiency, complemented by both case study analysis of Typhoon Hinnamnor and long-term statistical evaluations, the following main conclusions are as follows.

(1) The ARMS exhibits excellent accuracy and computational efficiency. It matches the accuracy of RTTOV11.2 in simulating satellite brightness temperatures, and after quality control, the data volume entering the assimilation system is similar to that of RTTOV, with an error margin of only 0.5%, while maintaining high computational speed.

(2) The use of the ARMS leads to improved assimilation and forecast outcomes. The forecast verification scores indicate a slightly positive effect on overall forecast performance when the ARMS is used as an observational operator in the YH4DVAR system, particularly in the Southern Hemisphere and tropical zone, where forecast accuracy is marginally enhanced.

(3) The ARMS can enhance the intensity forecast of Typhoon Hinnamnor. Compared with RTTOV, there is a certain improvement in typhoon intensity forecast when the ARMS is used, with a reduction of approximately 2% in the average typhoon wind speed error and approximately 1% in the average minimum pressure error.

Currently, this paper only investigates the application effects of the ARMS fast radiative transfer model in assimilating *MetOp-C* AMSU-A data into the YH4DVAR system. In future work, we plan to expand our research scope to include a variety of meteorological satellite instruments, such as the Advanced Technology Microwave Sounder (ATMS), MWTS, MWHS, Infrared Atmospheric Sounding Interferometer (IASI), and Hyperspectral Infrared Atmospheric Sounder (HIRAS), to fully test the application effects of the ARMS. Furthermore, given that the YH4DVAR system already has the all-sky assimilation capabilities for microwave humidity sounders (Ma et al., 2022), we further explore the integration of the ARMS scattering module into the YH4DVAR system to fully leverage the technical advantages of the ARMS vector radiative transfer model and implement an all-sky assimilation scheme based on the ARMS.

Owing to its outstanding accuracy, stability, and computational efficiency, the ARMS provides a new observational operator for the YH4DVAR system. The compatibility of the ARMS not only covers mainstream meteorological satellite detection instruments internationally but also fully supports the instruments of Chinese meteorological satellites while also having the ability to update relevant coefficients quickly. These features lay a solid foundation for the assimilation of Chinese meteorological satellite data. As our research on the application effects of ARMS in different meteorological satellite instruments deepens, it is expected to further enhance the assimilation capabilities and forecast accuracy of the YH4DVAR system, providing more reliable scientific support for global and regional weather forecasting.

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